FINANCIAL TRANSACTIONS & LOANS DASHBOARD

MICROSOFT POWER BI PROJECT REPORT

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TOOL USED: MICROSOFT POWER BI DESKTOP

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1. INTRODUCTION

For this project, I developed a Financial Transactions & Loans Dashboard using Microsoft Power BI. The project uses a banking dataset containing 11 tables related to customers, accounts, loans, transactions, branches and other details.

The main aim of this dashboard is to present raw data in a clear and interactive format. It helps users understand customer details, account activities, loan information and transaction performance through KPI cards, charts, tables and slicers.

2. OBJECTIVES

- To study customer, account, loan and transaction details.
- To calculate important indicators such as Total Customers, Active Accounts, Active Loans and Total Transaction Value.
- To compare different account types and loan statuses.
- To observe transaction patterns over different periods.
- To develop an interactive dashboard using slicers.
- To apply drill-down, tooltips and conditional formatting for better data understanding.

3. DATASET DESCRIPTION

The dataset used for this project consists of the following 11 tables:

- Customers
- Customer Types
- Accounts

- Account Types
- Account Statuses
- Loans
- Loan Statuses
- Transactions
- Transaction Types
- Branches
- Addresses

These tables provide information about customers, account balances, loan amounts, interest rates and transaction records. The tables are connected to support the analysis of financial activities.

4. DATA PREPARATION

Before designing the dashboard, I cleaned and formatted the dataset using Power Query Editor.

The following steps were performed:

- ❖ Removed duplicate values from Account ID, CustomerID, Address ID and LoanID colum
- ❖ Changed the date columns to the Date data type.
- ❖ Converted Balance, PrincipalAmount and Amount into Fixed Decimal Number format.
- ❖ Replaced missing customer names and address details with “unknown”.
- ❖ Corrected the DOB format using the English (India) locale and replaced errors with null values.
- ❖ Applied Trim and Clean transformations to remove unwanted spaces and characters from transaction descriptions.

These steps helped improve the quality and consistency of the data before analysis.

5. RELATIONSHIPS

I created 12 relationships between the tables in Model view. These relationships connect customer details with accounts, accounts with loans, and transactions with transaction types and branches.

Some relationships were kept inactive, including the relationship between Accounts and Loans and the relationship between Accounts and Transactions through AccountDestinationID.

The USERELATIONSHIP() function can be used in specific DAX calculations to work with these inactive relationships when required.

6. DAX MEASURES

I created the following DAX measures to calculate the important financial indicators used in the dashboard.

- ✓ **Total Customers:** Counts the unique customers in the dataset.
- ✓ **Total Accounts:** Calculates the number of distinct accounts.

- ✓ **Total Account Balance:** Adds the balances of all accounts.
- ✓ **Active Accounts:** Counts accounts that have Active status.
- ✓ **Total Loans:** Counts the distinct loan records.
- ✓ **Total Loan Principal:** Calculates the combined principal amount of loans.
- ✓ **Average Interest Rate:** Finds the average interest rate.
- ✓ **Active Loans:** Counts loans with Active status.
- ✓ **Total Transactions:** Counts the unique transaction IDs.
- ✓ **Total Transaction Value:** Calculates the sum of transaction amounts.

In addition, I created Destination Transaction Value and Total Loan Amount measures using USERELATIONSHIP() to analyse data through inactive relationships.

7. VISUALS USED

7.1 KPI CARDS

I created four KPI cards to display Total Customers, Active Accounts, Active Loans and Total Transaction Value. These cards provide a quick summary of the main financial information.

7.2 ACCOUNTS BY ACCOUNT TYPE

I used a Clustered Column Chart to compare the number of accounts under different account categories. Account Type was placed on the X-axis and Total Accounts on the Y-axis.

This chart helps identify differences in the number of accounts across categories.

7.3 TRANSACTIONS BY MONTH

I created a Line Chart using the Transaction Date hierarchy and Total Transactions. This visual helps track changes in transaction activity over time.

7.4 LOANS BY LOAN STATUS

I used a Donut Chart to show the distribution of loans according to their status. Loan Status was used as the category, and Total Loans was used as the value.

This chart makes it easier to compare loans across different statuses.

7.5 TRANSACTIONS BY TYPE

I created a Clustered Bar Chart using Transaction Type and Total Transactions. It helps compare the number of transactions recorded under each transaction category.

7.6 TRANSACTION SUMMARY

I used a Table visual to display Transaction Type, Total Transactions and Total Transaction Value.

This table provides detailed information and helps compare transaction counts and values across different categories.

8. SLICERS AND INTERACTIVITY

I added three slicers to make the dashboard more interactive.

- **Year:** Allows users to filter transactions according to the year.
- **Account Type:** Filters the report based on the account category.

- **Transaction Type:** Displays information for selected transaction categories.

I also used Format → Edit interactions to control the way visuals interact with one another. The available options include Filter, Highlight and None. This helps control how the selected data affects the other visuals.

9. DRILL-DOWN

I created a line chart titled “Transactions Over Time” using the Transaction Date hierarchy on the X-axis and transaction details on the Y-axis.

The drill-down feature was used to examine transaction activity at three levels: year, quarter and month.

- **Year Level:** The X-axis was set to Year, and the Y-axis range was adjusted from 0 to 13,000 to observe the overall transaction pattern.
- **Quarter Level:** I drilled down from year to quarter and set the Y-axis range from 11,500 to 12,500 to examine quarterly transaction activity.

- **Month Level:** I moved further down to the month level and adjusted the Y-axis range from 3,000 to 5,000 to study monthly patterns.

I also used Drill Down mode to examine a particular data point. I selected the year 2021, moved to Quarter 2 and then explored April, May and June.

This process helped me view transaction activity at different levels and study changes within a selected period.

I captured screenshots of the Year, Quarter, Month and selected data-point drill-down views to record the analysis.

10. REPORT-PAGE TOOLTIP

I created a separate report page named “Transaction Tooltip” and enabled the tooltip option in Page information.

Three cards were added to this page to display Transaction Type, Total Transactions and Total Transaction Value.

I then connected this tooltip page to the Transactions by Type chart. When users hover over a data point, the tooltip provides additional information about the selected transaction category.

11. CONDITIONAL FORMATTING

I applied data bars to the Total Transaction Value column in the Transaction Summary table.

As specified in the project instructions, green bars were used for positive values and red bars for negative values. This makes it easier to compare transaction values visually.

12. KEY FINDINGS

The dashboard helps examine the following areas:

- The total number of customers and accounts.
- The number of accounts and loans that are currently active.
- The account category with the highest number of accounts.
- Changes in transaction counts over time.

- The distribution of loans across different statuses.
- Differences in transaction counts and values among transaction categories.

The actual numerical findings should be added after checking and validating the results in Power BI.

CONCLUSION

Through this project, I gained practical knowledge of data cleaning, relationship creation, DAX calculations and dashboard development using Microsoft Power BI.

The KPI cards, charts, tables and slicers help present financial information in an organised and interactive way. They also make it easier to examine customer details, account activity, loan information and transaction performance.

Overall, this project improved my understanding of data modelling, financial data analysis and the development of interactive dashboards using Microsoft Power BI.